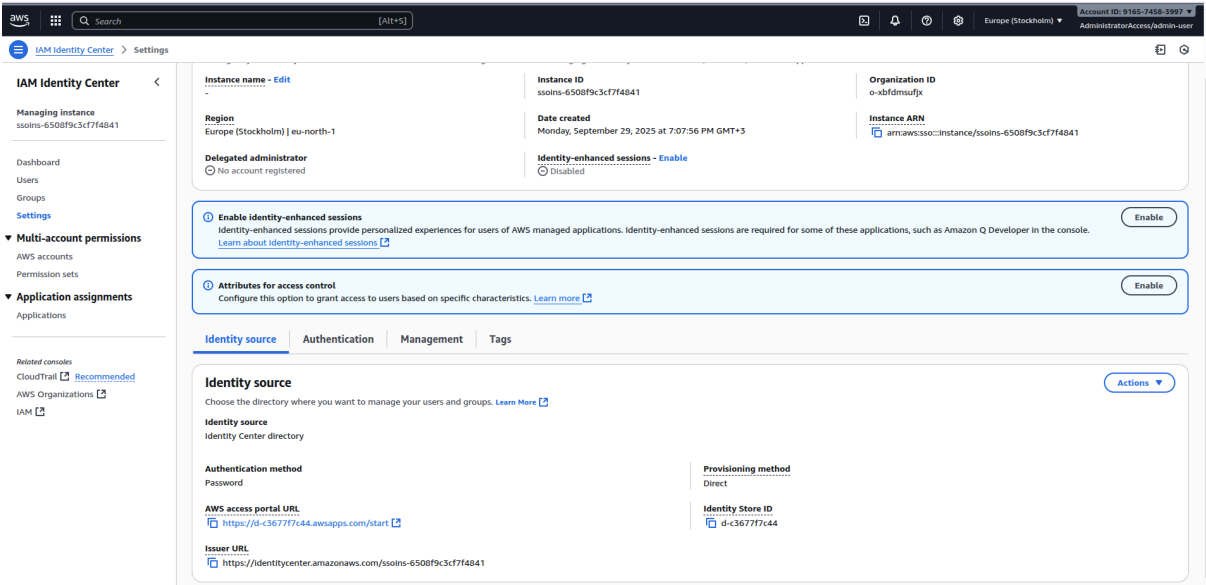


1. Created AWS account
2. IAM identity center account with full-admin user



Select permission set type

A permission set contains policies that determine a user's permissions to access an AWS account. When you assign a user or group to a permission set in an AWS account, IAM Identity Center creates an IAM role in the account and attaches the policies specified in the permission set to that role. Select an option to specify the permission set type. [Learn more](#)

Permission set type

Types

☒ Predefined permission set

Create a predefined permission set by choosing an AWS-defined template. This template enables you to select a single AWS managed policy. For example, you can select a policy that grants permissions for a common job function, such as Billing, or a specific level of access to AWS services and resources, such as ViewOnlyAccess. You can update the permission set as your needs evolve.

☐ Custom permission set

Create a custom permission set by selecting AWS managed policies and creating an inline policy (recommended). You can also attach customer managed policies and set a permissions boundary (advanced).

Policy for predefined permission set

Select an AWS managed policy

☒ AdministratorAccess

Provides full access to AWS services and resources.

Permission sets (1)

Permission sets define the level of access that users in IAM Identity Center have to their assigned AWS accounts. The names of permission sets appear as available roles in the AWS access portal. To sign in to the AWS access portal, choose an account, and then choose a role that AWS created from an assigned permission set. [Learn more](#)

Search permission sets by name, ARN, or ID

Permission set	Description	ARN
☐ AdministratorAccess	-	arn:aws:sso::permissionSet/ssoins-6508f9c3cf7f4841/ps-c096f58bb0e7...

Select permission sets

Assign permission sets to "artem_polishchuk4315"

Permission sets define the level of access that users and groups in IAM Identity Center have to an AWS account. You can assign more than one permission set to a user. To ensure least privilege access to AWS accounts, users in IAM Identity Center with multiple permission sets on an AWS account must pick a specific permission set when selecting the account and then return to the AWS access portal to pick a different set when necessary [Learn more](#)

Permission sets (1/1)

Search permission sets by name, ARN, or ID (i.e., ps-abcdefg123456789)

1

<

>

☒	Permission set	Description	ARN
☒	AdministratorAccess	-	arn:aws:sso::permissionSet/ssoins-6508f9c3cf7f4841/ps-c096f58bb0e75296

Cancel

Previous

Next

Account ID: 9165-7458-3997
AdministratorAccess/admin-user

3. Organisation

AWS Organizations

AWS accounts

AWS Organizations

AWS accounts

Invitations

Multi-party approval [New](#)

Services

Policies

Settings [New](#)

Get started

Organization ID
o-xbfdm5ufjx

AWS accounts

Add an AWS account

The accounts listed below are members of your organization. The organization's management account is responsible for paying the bills for all accounts in the organization. You can use the tools provided by AWS Organizations to centrally manage these accounts. [Learn more](#)

Centralize root access for member accounts

You can delete root credentials for your member accounts and perform privileged actions from the management or delegated account. [Learn more about centralizing root access](#)

Enable in IAM

Organization

Organizational units (OUs) enable you to group several accounts together and administer them as a single unit instead of one at a time.

Search by name, email, account ID or OU ID.

Hierarchy

List

Organizational structure

Account created/joined date

Root

r-8E3p

artem_polishchuk4315

management account

916574583997 | polishchuk.artem@kpl.ua

Joined 2025/05/29

4. Add view only user and policy

Users (2)

Delete users

Add user

Users listed here can sign in to the AWS access portal to access AWS accounts and assigned cloud applications. [Learn more](#)

Username

Search users

<

1

>

☐	Username	Display name	Status	MFA devices	Created by
☐	admin-user	Artem Polishchuk	Enabled	1 device	Manual
☐	view-only-user	view-only-user	Enabled	None	Manual

Creating policy that allows only switch to role

AllowSwitchToViewOnlyRolePolicy Info Edit Delete

Policy details

Type
Customer managed

Creation time
September 29, 2025, 20:13 (UTC+03:00)

Edited time
September 29, 2025, 20:13 (UTC+03:00)

ARN
 am:aws:iam::916574583997:policy/AllowSwitchToViewOnlyRolePolicy

Permissions

Entities attached

Tags

Policy versions
(1)

Last Accessed

Permissions defined in this policy Info Copy Edit Summary JSON

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM Identity (user, user group, or role), attach a policy to it

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "sts:AssumeRole",
7       "Resource": "arn:aws:iam::916574583997:role/ViewOnlyRole"
8     }
9   ]
10 }
```

Creating permission set that allows to attach AllowSwitchToViewOnlyRolePolicy

AWS managed policies (0) Detach Attach policies

AWS managed policies are standalone policies that are created and managed by AWS. Different types of AWS managed policies enable you to grant predefined permissions for many common use cases. For example, you can use job function policies to grant permissions for common job functions, full access policies to grant service administrators full access to an AWS service, and partial access policies to grant specific levels of access to AWS services. You can select up to 10 managed policies (AWS managed policies and customer managed policies) for your permission set. [Learn more](#)

< 1 > ⚙

Policy name	Type	Description
No AWS managed policies attached		
Attach policies		

Customer managed policies (1) Detach Attach policies

Customer managed policies are standalone policies that you create and manage in your AWS accounts to define custom permissions. You can attach up to 10 managed policies (AWS managed policies and customer managed policies) to your permission set by specifying the names of the policies exactly as they appear in your accounts. Customer managed policies are intended for advanced use cases. To ensure that you understand your shared security responsibility and best practices for configuring these policies, review the IAM Identity Center documentation. [Learn more](#)

< 1 > ⚙

Policy name
☐ AllowSwitchToViewOnlyRolePolicy

Assign permission set to the view only user

Select permission sets

Assign permission sets to "artem_polishchuk4315"

Permission sets define the level of access that users and groups in IAM Identity Center have to an AWS account. You can assign more than one permission set to a user. To ensure least privilege access to AWS accounts, users in IAM Identity Center with multiple permission sets on an AWS account must pick a specific permission set when selecting the account and then return to the AWS access portal to pick a different set when necessary. [Learn more](#)

Permission sets (1/2) ⌂ Create permission set

< 1 > ⚙

Permission set	Description	ARN
☐ AdministratorAccess	-	arn:aws:sso::permissionSet/ssolns-6508f9c3cf7f4841/ps-c096f58bb0e75296
☒ ViewOnlySwitchAccess	-	arn:aws:sso::permissionSet/ssolns-6508f9c3cf7f4841/ps-a2bd949c77a7afe4

Cancel Previous Next

Switching to role

Switch Role

Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. [Learn more](#)

Account ID

The 12-digit account number or the alias of the account in which the role exists.

IAM role name

The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the **TestRole** role name from the following role ARN: `arn:aws:iam::123456789012:role/TestRole`.

Display name - optional

This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.

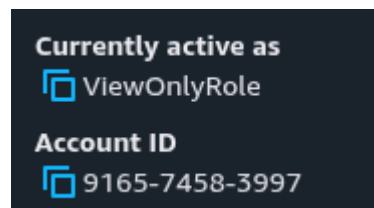
Display color - optional

The selected color displays in the console navigation when this role is active

☐ None

Cancel

Switch Role



6. Creation of policy and role via Terraform

Created [main.tf](#) and terraform.tfstate files.

<https://github.com/CTS2025-kpi/lab-1-infrastructure-setup-artemp4315/pull/2/files>

Ran terraform apply

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

ViewOnlyRole-Terraform

Info

Delete

Summary

Edit

Creation date

September 30, 2025, 15:14 (UTC+03:00)

ARN

arn:aws:iam::916574583997:role/ViewOnlyRole-Terraform

Link to switch roles in console

https://signin.aws.amazon.com/switchrole?roleName=ViewOnlyRole-Terraform&account=916574583997

Last activity

-

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1)

Info

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Filter by Type

All types

Search

Policy name

Type

Attached entities

CustomViewOnlyPolicy-Terraform

Customer managed

1

CustomViewOnlyPolicy-Terraform

Info

Edit

Delete

A custom view-only policy created by Terraform.

Policy details

Type

Customer managed

Creation time

September 30, 2025, 15:14 (UTC+03:00)

Edited time

September 30, 2025, 15:14 (UTC+03:00)

ARN

arn:aws:iam::916574583997:policy/CustomViewOnlyPolicy-Terraform

Permissions

Entities attached

Tags

Policy versions

Last Accessed

Permissions defined in this policy

Info

Copy

Edit

Summary

JSON

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

```

1- {
2-   "Statement": [
3-     {
4-       "Action": [
5-         "ec2:Describe*",
6-         "s3:List*",
7-         "rds:Describe*"
8-       ],
9-       "Effect": "Allow",
10-      "Resource": "*"
11-    }
12-  ],
13-   "Version": "2012-10-17"
14- }

```

7. Create a network (VPC for your resources)

Created [network.tf](#)

<https://github.com/CTS2025-kpi/lab-1-infrastructure-setup-artemp4315/pull/2/files>

terraform plan

terraform apply

Apply complete! Resources: 6 added, 0 changed, 0 destroyed.

VPC

Your VPCs (2)

Info

Last updated less than a minute ago

Actions

Create VPC

Find VPCs by attribute or tag

Name

VPC ID

State

Block Public...

IPv4 CIDR

IPv6 CIDR

DHCP option set

Main route table

custom-vpc

vpc-068026a409d15479f

Available

Off

10.0.0.0/22

-

dopt-0080f497c597864...

rtb-06bb4ee4fc48

-

vpc-0a820fa5769ee7a41

Available

Off

172.31.0.0/16

-

dopt-0080f497c597864...

rtb-0ec240d5b4ec

Subnets

Subnets (5) Info

Last updated 1 minute ago

Actions

Create subnet

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIE
☐	public-subnet-1	subnet-0e6fdbfa6819b0a56	Available	vpc-068026a409d15479f cust...	Off	10.0.1.0/24	-	-
☐	-	subnet-0e015d51569f2ecbb	Available	vpc-0a870fa5769ee7a41	Off	172.31.0.0/20	-	-
☐	private-subnet-1	subnet-0124f1878128dcc19	Available	vpc-068026a409d15479f cust...	Off	10.0.2.0/24	-	-
☐	-	subnet-03394a93ba7d7f5d5	Available	vpc-0a870fa5769ee7a41	Off	172.31.16.0/20	-	-
☐	-	subnet-0de99431710facc67	Available	vpc-0a870fa5769ee7a41	Off	172.31.32.0/20	-	-

IGW

Internet gateways (2) Info

Last updated 2 minutes ago

Actions

Create internet gateway

Find Internet gateways by attribute or tag

☐	Name	Internet gateway ID	State	VPC ID	Owner
☐	-	igw-001e04df540f0a338	Attached	vpc-0a870fa5769ee7a41	916574583997
☐	custom-igw	igw-0ed1242ba57d3fecf	Attached	vpc-068026a409d15479f custom-vpc	916574583997

Route table

Route tables (3) Info

Last updated 2 minutes ago

Action

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	public-route-table	rtb-0d720a88781a8a8a0	subnet-0e6fdbfa6819b0a56...	-	No	vpc-068026a409d15479f cust...	916574583997

rtb-0d720a88781a8a8a0 / public-route-table

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-0ed1242ba57d3fecf	Active	No
10.0.0.0/22	local	Active	No

rtb-0d720a88781a8a8a0 / public-route-table

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (1)

Find subnet association

Edit subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
public-subnet-1	subnet-0e6fdbfa6819b0a56	10.0.1.0/24	-

Subnets without explicit associations (1)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
private-subnet-1	subnet-0124f1878128dcc19	10.0.2.0/24	-

8. IaC stack

<https://github.com/CTS2025-kpi/lab-1-infrastructure-setup-artemp4315/pull/2/files>

9. Calculations

AWS Fargate

Fargate Spot and Compute Savings Plans are currently not supported on the AWS Pricing Calculator.

With AWS Fargate, there are no upfront payments and you only pay for the resources that you use. Pricing is based on requested vCPU, memory, and storage resources consumed by your Amazon ECS Task or Amazon EKS Pod.

Operating system

Linux

CPU Architecture

x86

Number of tasks or pods

Enter the number of tasks or pods running for your application

Value

3

Unit

per month

Average duration

Enter the time period for which your tasks or pods are running. Pricing is per second with a 1-minute minimum (Linux) and a 5-minute minimum (Windows). Duration is calculated from the time you start to download your container image (docker pull) until the Task or Pod terminates, rounded up to the

Value

730

Unit

hours

Amount of vCPU allocated

Enter the amount between 0.25 vCPU and 16 vCPU

0.5

vCPU selected supports memory values between 1 GB and 4 GB, in 1 GB increments

Amount of memory allocated

Unit

1

GB

Total Upfront cost: 0.00 USD

Total Monthly cost: 59.46 USD

Show Details

ALB

▼ Show calculations

0.20 GB per hour / 0.4 GB processed bytes per hour per LCU for Lambda functions as targets = 0.50 processed bytes LCUs for Lambda functions as targets

0.50 GB per hour / 1 GB processed bytes per hour per LCU for EC2 Instances and IP addresses as targets = 0.50 processed bytes LCUs for EC2 Instances and IP addresses as targets

0.50 LCUs + 0.50 LCUs = 1 processed bytes LCUs

2 new connections per second / 25 new connections per second per LCU = 0.08 new connections LCUs

2 new connections per second x 2 seconds = 4 active connections

4 active connections / 3000 connections per LCU = 0.0013333333333333333 active connections LCUs

0 rules per request - 10 free rules = -10 paid rules per request after 10 free rules

Max (-10 USD, 0 USD) = 0.00 paid rules per request

Max (1 processed bytes LCUs , 0.08 new connections LCUs , 0.0013333333333333333 active connections LCUs , 0 rule evaluation LCUs) = 1 maximum LCUs

1 load balancers x 1 LCUs x 0.0076 LCU price per hour x 730 hours per month = 5.55 USD

Application Load Balancer LCU usage charges (monthly): 5.55 USD

Total Upfront cost: 0.00 USD

Total Monthly cost: 23.03 USD

Show Details

Cancel

Amazon API Gateway

Step 1

Add service

Step 2

Configure service

Create estimate: Configure Amazon API Gateway

Info

Description

Enter a description for your estimate

Choose a location type

Info

Region

Choose a Region

Europe (Stockholm)

HTTP APIs

Info

Select the units, number, and frequency for HTTP API requests based on expected volume. The calculations below exclude free tier discounts.

HTTP API requests units

The unit will apply to the number of requests you will specify below.

exact number

Requests

5000

Unit

per month

Average size of each request

Requests are defined as having 512 KB of data; a request with 513 KB of data is metered as two requests.

Value

34

Unit

KB

► Show calculations

Total Upfront cost: 0.00 USD

Total Monthly cost: 0.01 USD

Show Details

Cancel

Save and view summary

DynamoDB

Choose DynamoDB features

Choose the DynamoDB features whose pricing you want to estimate.

☒ DynamoDB on-demand capacity

☐ DynamoDB Streams

☐ DynamoDB Data export to Amazon S3

☐ DynamoDB provisioned capacity

☐ DynamoDB Backup and restore

☐ DynamoDB Data import from Amazon S3

☐ DynamoDB Accelerator (DAX) clusters

☐ DynamoDB change data capture

DynamoDB on-demand capacity feature

Table Class

Info

DynamoDB offers two table classes designed to help you optimize for cost. The DynamoDB Standard table class is the default and recommended for the vast majority of workloads. The DynamoDB Standard-infrequent Access (DynamoDB Standard-IA) table class is optimized for tables that store data that is accessed infrequently, where storage is the dominant cost. Each table class offers different pricing for data storage as well as read and write requests.

Table class

Select table class

Standard

Data storage

The calculations in this section exclude AWS Free Tier discounts.

Data storage size

Unit

Total Upfront cost: 0.00 USD

Total Monthly cost: 1.34 USD

Show Details

Cancel

Update

Result

[AWS Pricing Calculator](#)

>

My Estimate

My Estimate

Edit

Estimate summary

Info

Upfront cost	Monthly cost	Total 12 months cost
0.00 USD	83.84 USD	1,006.08 USD
		Includes upfront cost